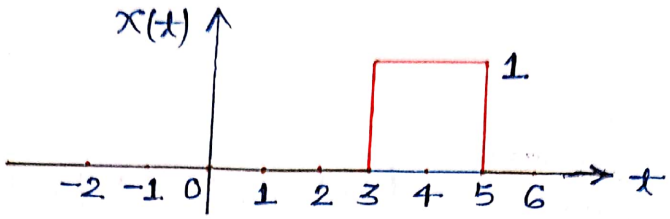
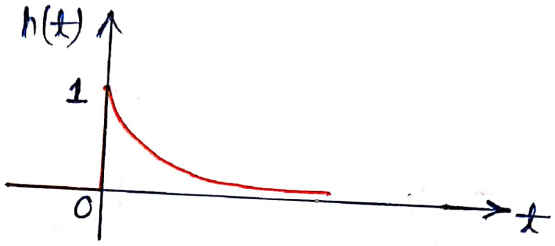
S&S QUIZ-3 SOLUTION

Sol(1):- Given- $x(t) = u(t-3) - u(t-5)$



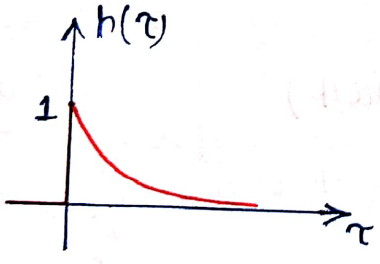
$h(t) = e^{-3t} \cdot u(t)$



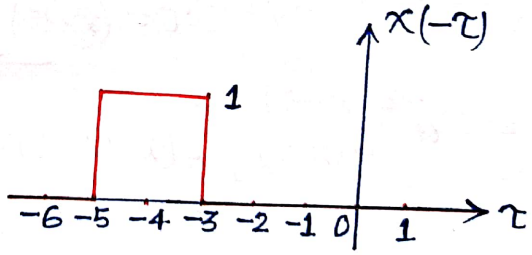
Part(a):- $y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) \cdot h(t-\tau)\, d\tau = \int_{-\infty}^{\infty} h(\tau) \cdot x(t-\tau)\, d\tau$

→ (0.5 Mark)

$$= \int_{-\infty}^{\infty} e^{-3\tau} u(\tau) \cdot \{u(t-\tau-3) - u(t-\tau-5)\}\, d\tau$$



→ (0.25 Mark)



→ (0.25 Mark)

Case(I):-

For $t \leq 3$, $y(t) = 0$ → (1 Mark)

Case(II):-

For $3 < t \leq 5$, $y(t) = \int_0^{t-3} e^{-3\tau}\, d\tau = \left\{\frac{e^{-3\tau}}{-3}\right\}_0^{t-3} = \left(\frac{1 - e^{-3(t-3)}}{3}\right)$

→ (1 Mark)

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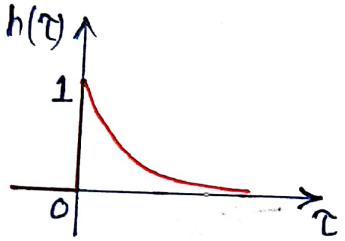
Case(III):-

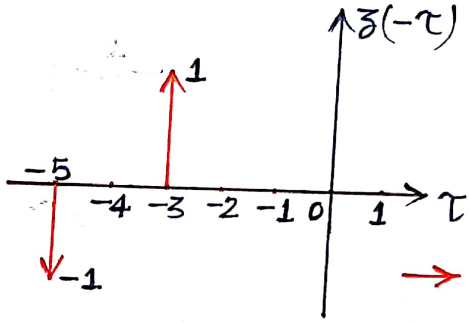
For $t>5$, $y(t)=\int_{t-5}^{t-3} e^{-3\tau}\,d\tau = \left\{\frac{e^{-3\tau}}{-3}\right\}_{t-5}^{t-3} = \left(\frac{e^{-3(t-5)} - e^{-3(t-3)}}{3}\right)$

→ (1 Mark)

Part(b):- $g(t)=\left\{\frac{dx(t)}{dt}\right\} * h(t) = z(t) * h(t)$

$= \int_{-\infty}^{\infty} z(\tau)\cdot h(t-\tau)\,d\tau = \int_{-\infty}^{\infty} h(\tau)\cdot z(t-\tau)\,d\tau$ → (0.5 Mark)





→ (0.5 Mark)

$\therefore g(t) = \int_{-\infty}^{\infty} \left\{e^{-3\tau}\cdot u(\tau)\right\}\left\{-\delta(t-\tau-5)+\delta(t-\tau-3)\right\} d\tau$

$= -\int_{-\infty}^{\infty} e^{-3\tau}\cdot \underbrace{\delta(t-\tau-5)}\cdot u(\tau)\,d\tau + \int_{-\infty}^{\infty} e^{-3\tau}\cdot \underbrace{\delta(t-\tau-3)}\,u(\tau)\,d\tau$

$t-\tau-5=0$ → (0.5 Mark)

$\tau=(t-5)$

$t-\tau-3=0$

$\tau=(t-3)$ → (0.5 Mark)

$= -e^{-3(t-5)}\cdot u(t-5)\int_{-\infty}^{\infty}\delta(t-\tau-5)\cdot d\tau + e^{-3(t-3)}\,u(t-3)\int_{-\infty}^{\infty}\delta(t-\tau-3)\,d\tau$

$= -e^{-3(t-5)}\,u(t-5) + e^{-3(t-3)}\cdot u(t-3)$

$\int_{-\infty}^{\infty}\delta(t-\tau-5)\,d\tau = 1$

$\int_{-\infty}^{\infty}\delta(t-\tau-3)\,d\tau = 1$ → (2 X 0.5 Mark)

$g(t) = e^{-3(t-3)}\cdot u(t-3) - e^{-3(t-5)}\cdot u(t-5)$ → (1 Mark)

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Part (c):-

As we know from Part (a),

$$y(t) = \begin{cases} 0, & t \leq 3 \\ \left(\frac{1 - e^{-3(t-3)}}{3}\right), & 3 < t \leq 5 \\ \left(\frac{e^{-3(t-5)} - e^{-3(t-3)}}{3}\right), & 5 < t \end{cases}$$

$$\frac{d}{dt}y(t) = \begin{cases} 0, & t \leq 3 \\ e^{-3(t-3)}, & 3 < t \leq 5 \\ e^{-3(t-3)} - e^{-3(t-5)}, & 5 < t \end{cases}$$

$$\therefore \frac{d}{dt}y(t) = e^{-3(t-3)}\{u(t-3) - u(t-5)\} + \{e^{-3(t-3)} - e^{-3(t-5)}\}u(t-5)$$

$$= e^{-3(t-3)} \cdot u(t-3) - e^{-3(t-5)} \cdot u(t-5)$$

$$= g(t)$$ → (2 Mark)

Hence, $g(t) = \frac{d}{dt}y(t)$

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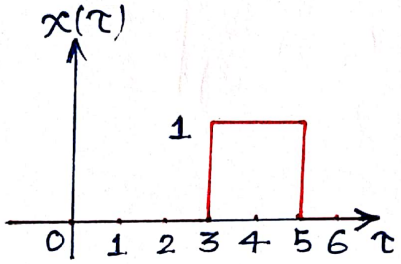
Alternate Method :-

Sol 1(a) :-

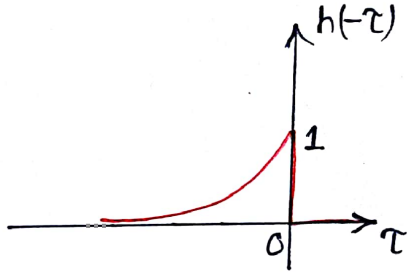
$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) \cdot h(t-\tau)\, d\tau$$

$$= \int_{-\infty}^{\infty} \{u(\tau-3) - u(\tau-5)\} \cdot e^{-3(t-\tau)} \cdot u(t-\tau)\, d\tau$$

→ (0.5 Mark)



→ (0.25 Mark)



→ (0.25 Mark)

Case (I) :- For $t \leq 3$, $y(t) = 0$ → (1 Mark)

Case (II) :- For $3 < t \leq 5$, $y(t) = \int_3^t e^{-3(t-\tau)}\, d\tau = \left\{ \frac{e^{-3(t-\tau)}}{3} \right\}_3^t$

$$= \left(\frac{1 - e^{-3(t-3)}}{3} \right)$$ → (1 Mark)

Case (III) :- For $t > 5$, $y(t) = \int_3^5 e^{-3(t-\tau)}\, d\tau = \left\{ \frac{e^{-3(t-\tau)}}{3} \right\}_3^5$

$$= \left(\frac{e^{-3(t-5)} - e^{-3(t-3)}}{3} \right)$$

→ (1 Mark)

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